Digital Assignment Lab - 2

Course Name :- Agile Development Process and Devops Lab

Course code :- ISWE406P

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1. Learning Goal

Understand **what Jenkins is** and run a job successfully.

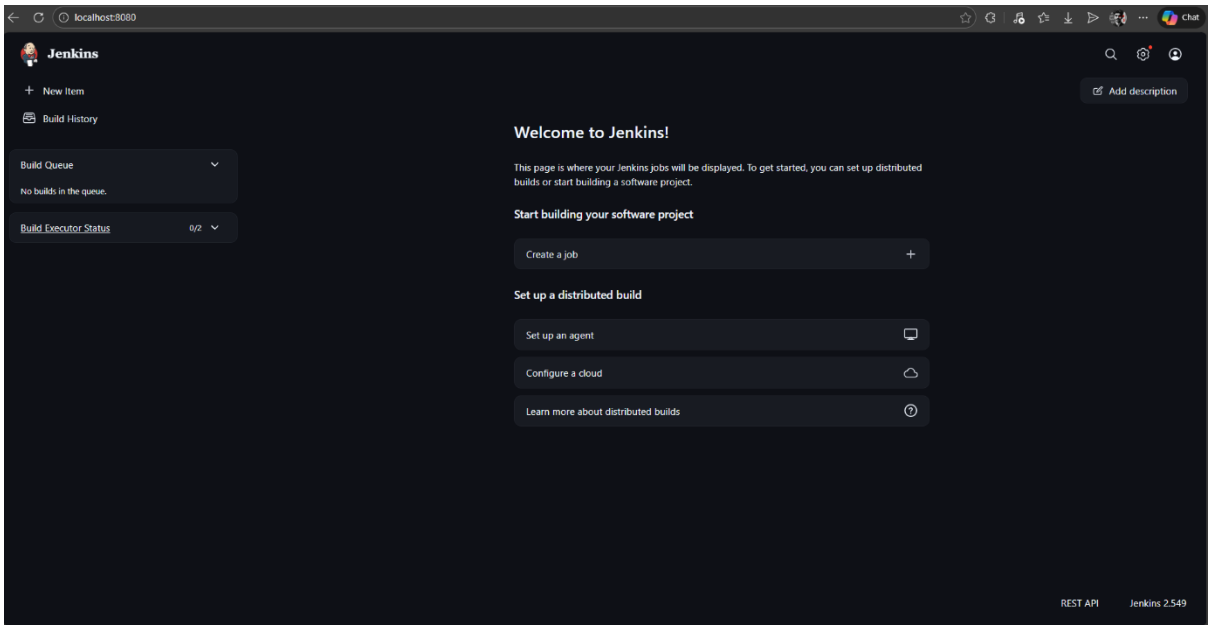
Stepwise Tasks

1. What is CI/CD? (Real-world example)

CI/CD stands for Continuous Integration and Continuous Delivery/Deployment.

- **Continuous Integration (CI):**
  Developers frequently push code to a shared repository (GitHub/GitLab). Jenkins automatically:
  - Pulls the latest code
  - Builds the project
  - Runs unit tests
  - Reports success or failure
- **Continuous Delivery/Deployment (CD):**
  After successful build and tests, Jenkins automatically deploys the application to:
  - Test server
  - Staging server
  - Production server

2. Install Jenkins (Windows / Linux)



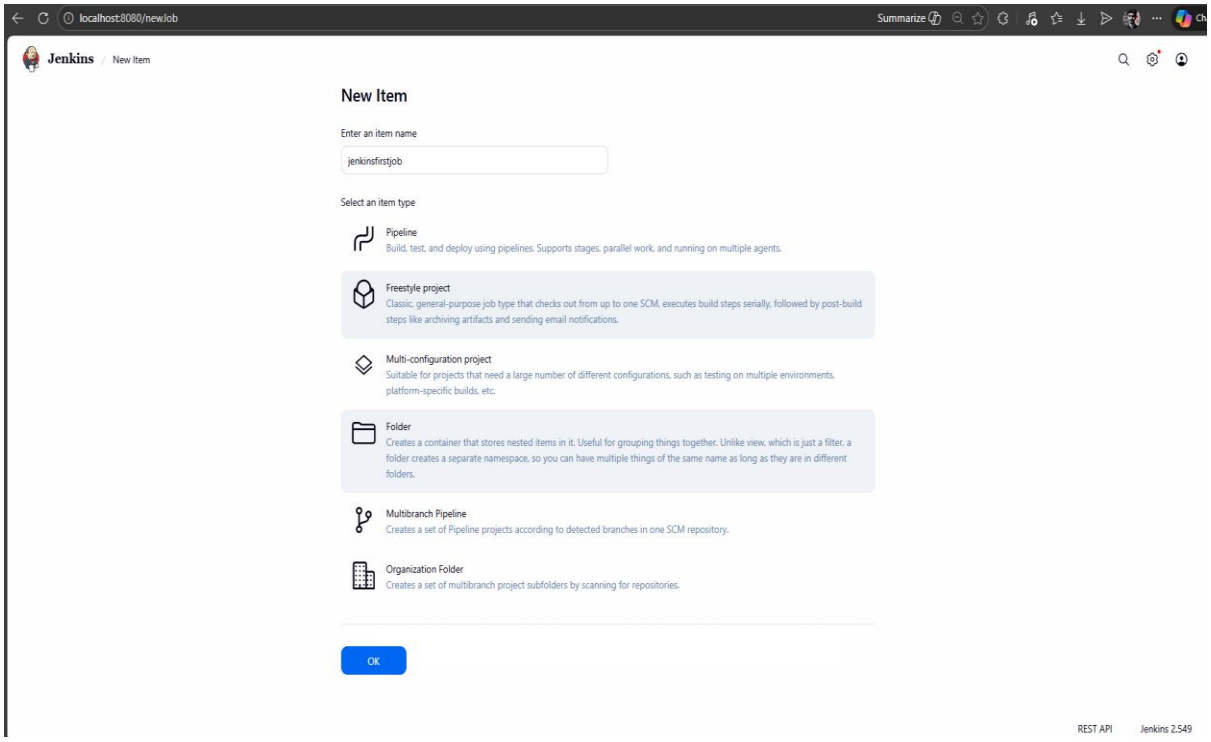
3. Jenkins UI overview

Main Components Visible in Jenkins UI

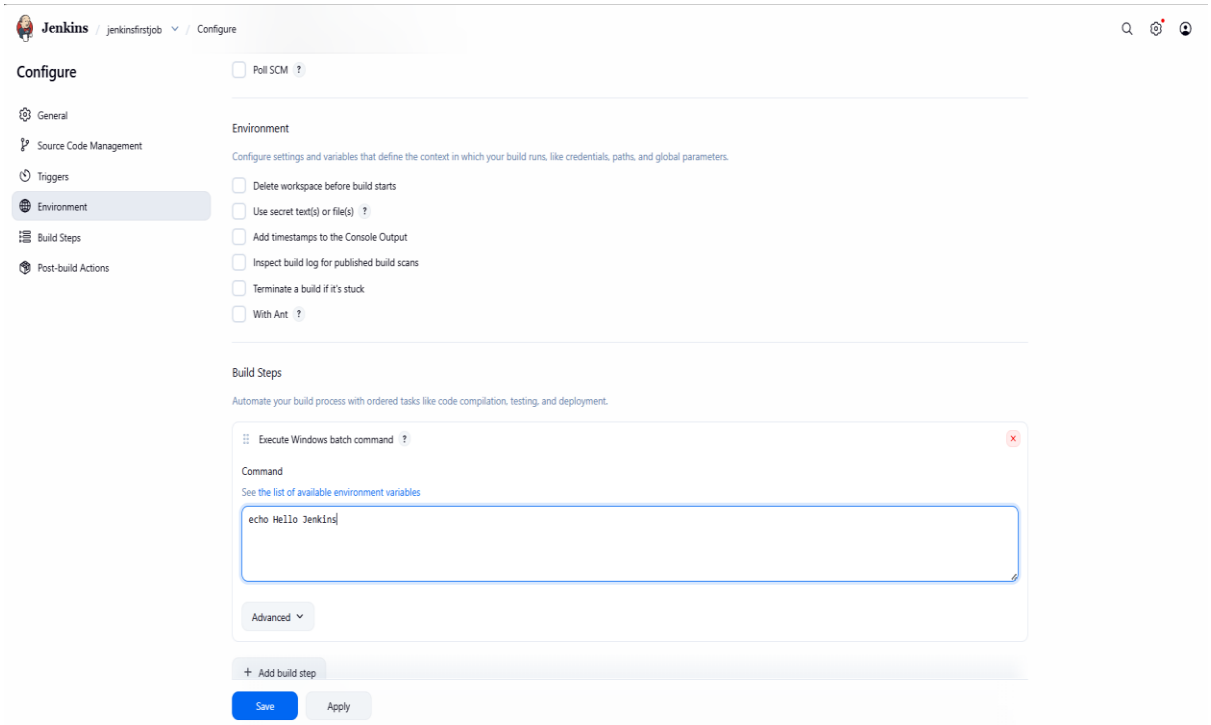
1. **Welcome Dashboard**
   - Displays the message *"Welcome to Jenkins!"*
   - Central area shows options to start creating jobs and configure agents.

2. **New Item**
    - Used to create:
        - Freestyle project
        - Pipeline job
        - Multibranch pipeline
    - This is the starting point for any automation task.
3. **Build History**
    - Shows the list of previously executed builds.
    - Helps to track:
        - Success
        - Failures
        - Build numbers and timestamps.
4. **Build Queue**
    - Displays jobs that are waiting to be executed.
    - Useful when multiple builds are triggered at the same time.
5. **Build Executor Status**
    - Shows how many executors (agents) are available and busy.
    - Helps monitor Jenkins performance.
6. **Create a Job**
    - Quick option in the center panel to create a new project.
7. **Set up an Agent / Cloud**
    - Used to configure distributed builds
    - Jenkins can run jobs on remote machines (agents).
8. **Manage Jenkins (Settings Icon)**
    - Used for:
        - Plugin installation
        - User management
        - Security settings
        - Tool configuration.
9. **Search & User Profile**
    - Top-right corner provides:
        - Search option
        - Logged-in user details
        - System configuration.

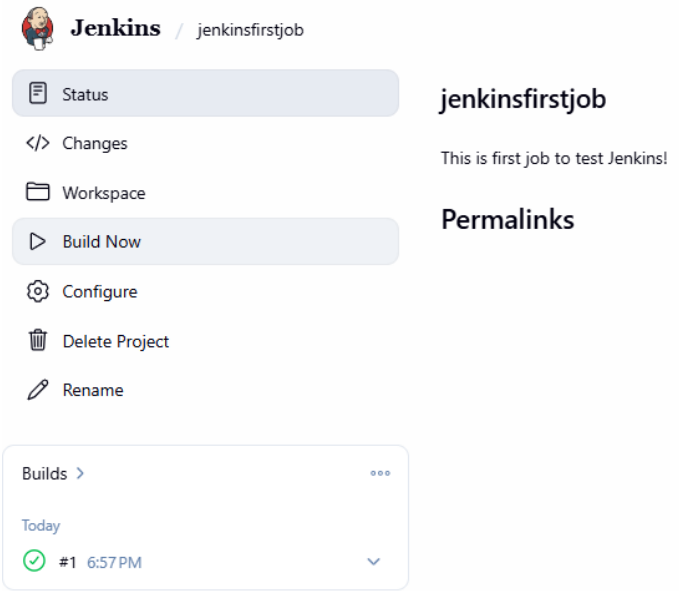
4. Create **Freestyle Job**



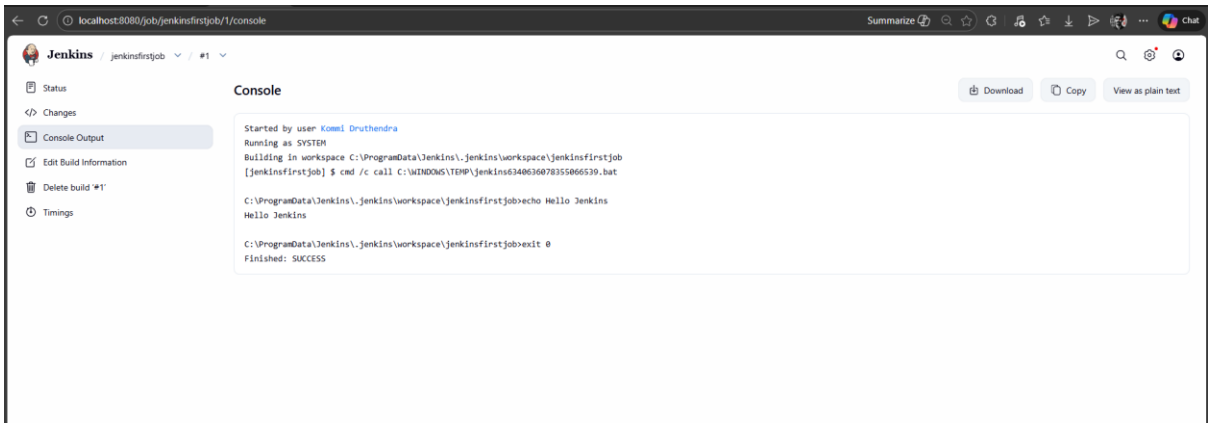
5. Add simple build step:
   - Windows: `echo Hello Jenkins`



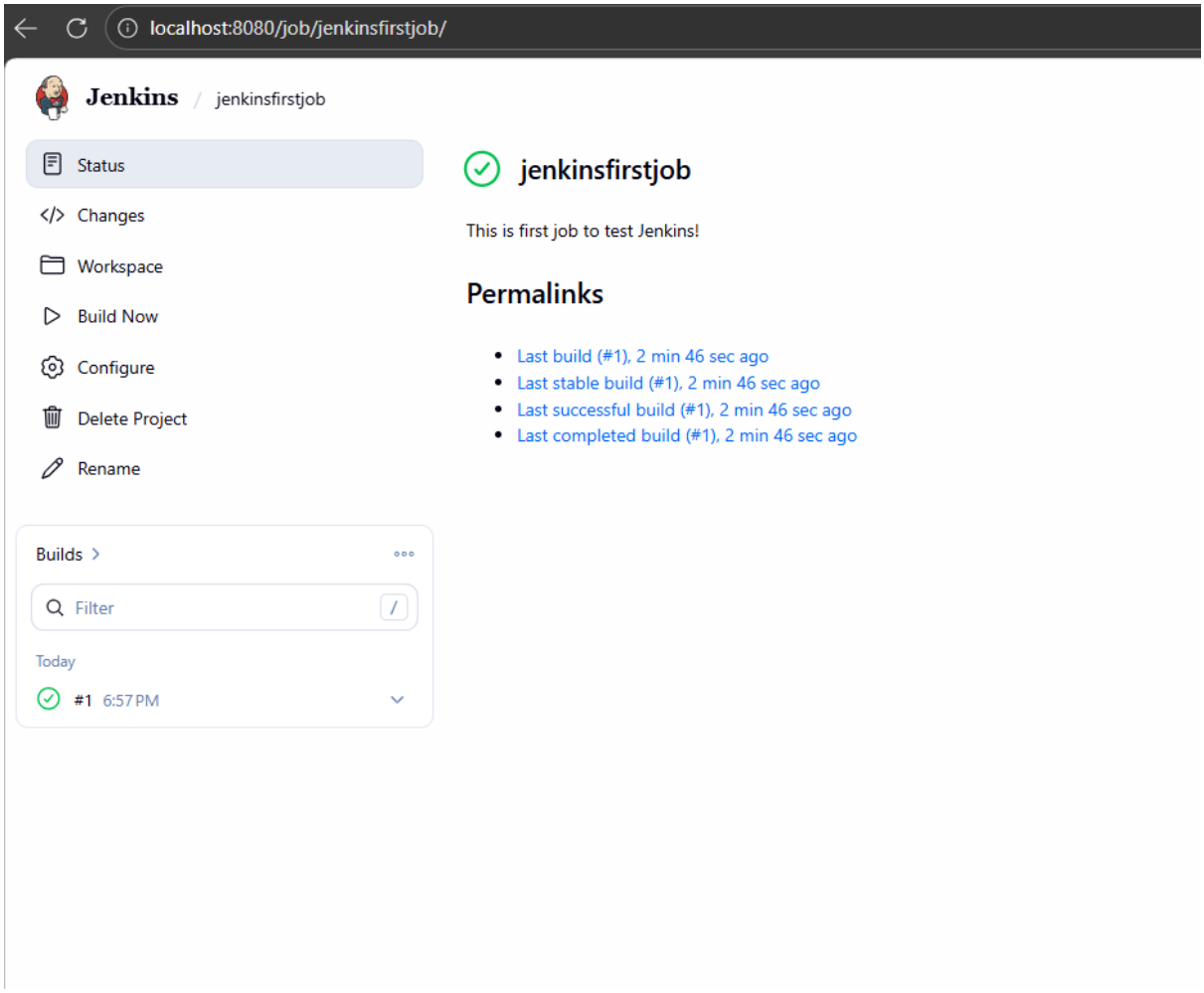
6. Run job manually



7. View:
   - Console Output



- Build History



Outcome

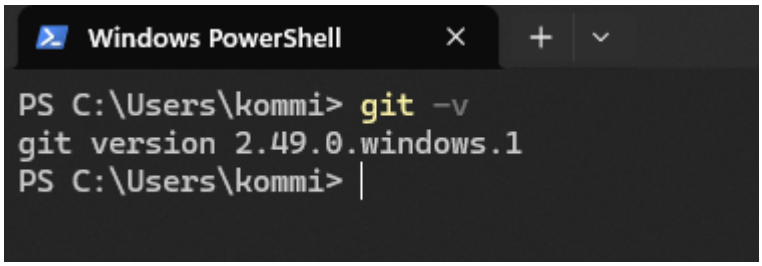
A learner can install Jenkins and run a basic job.

2. Learning Goal

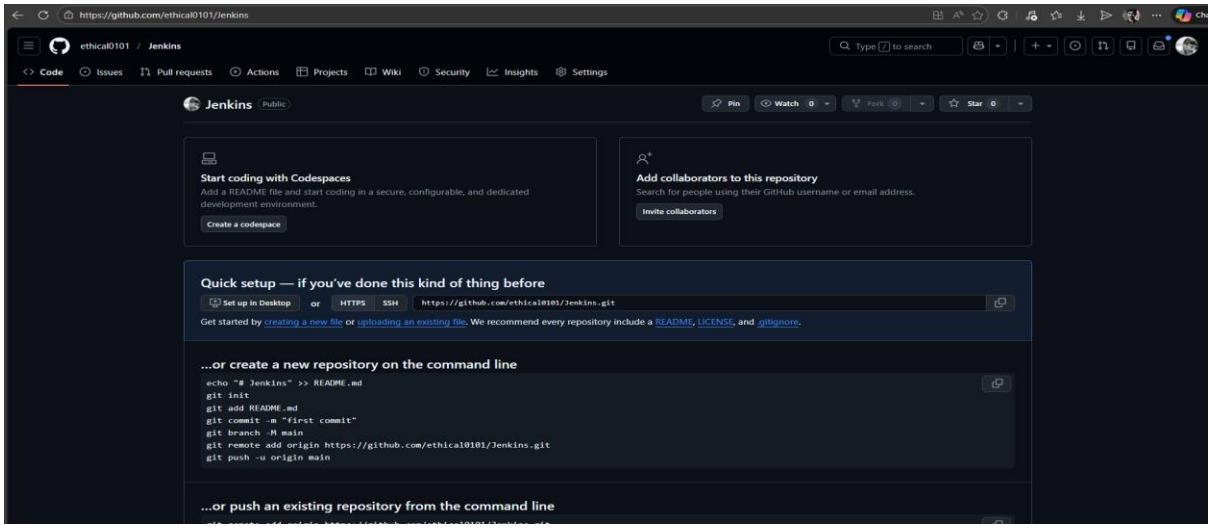
Work with **source code and build automation**

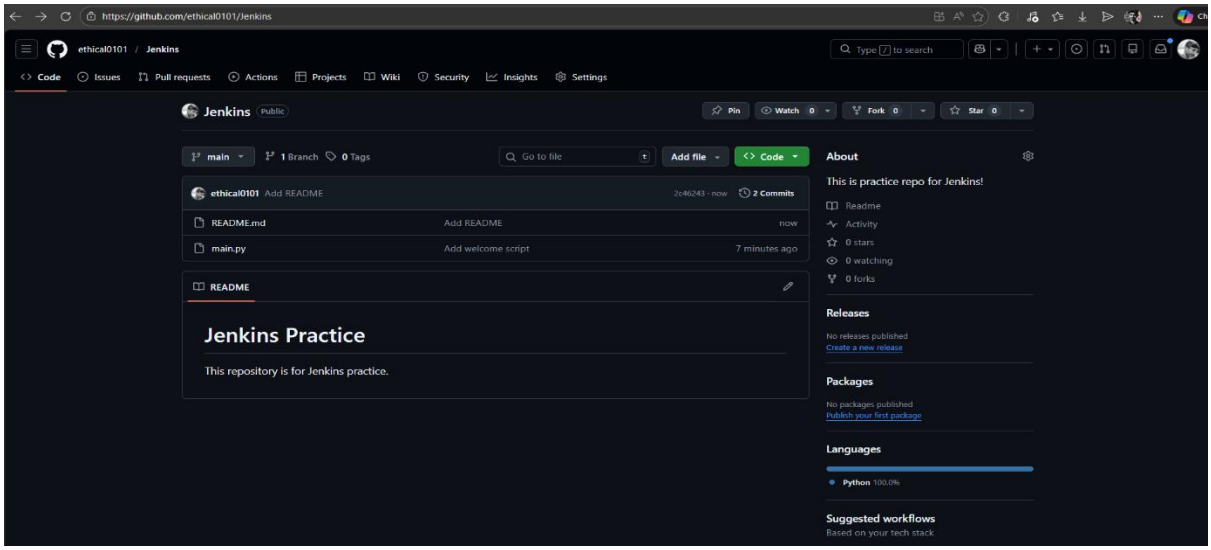
Stepwise Tasks

1. Install Git



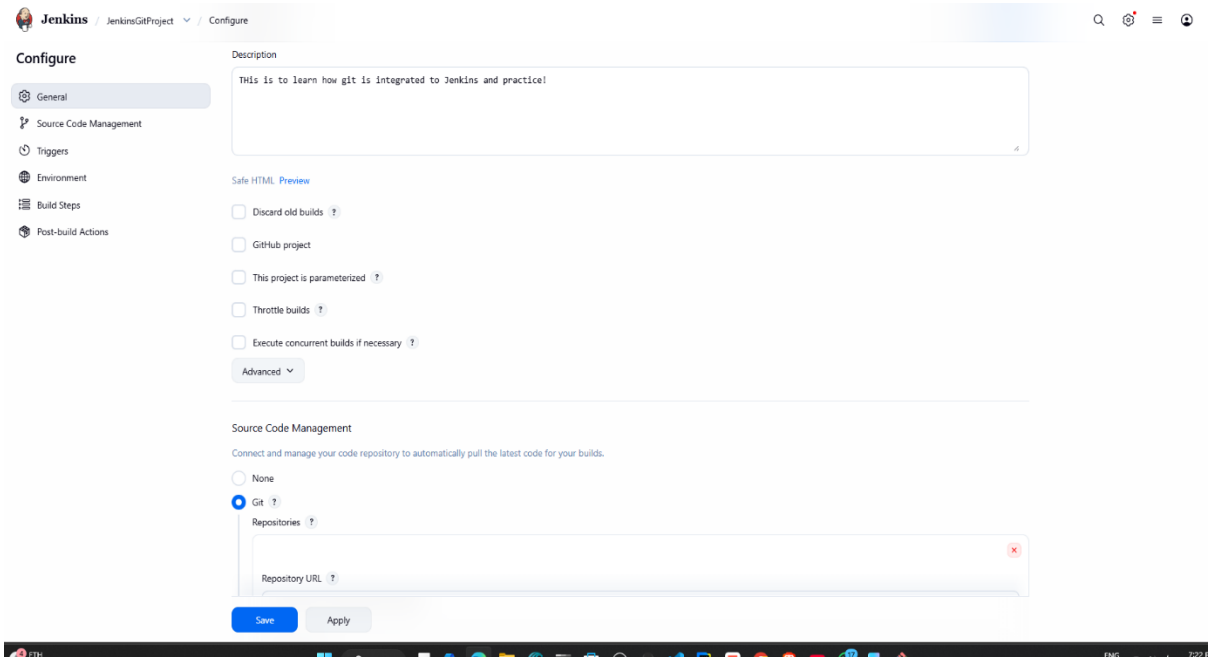
2. Create GitHub repository



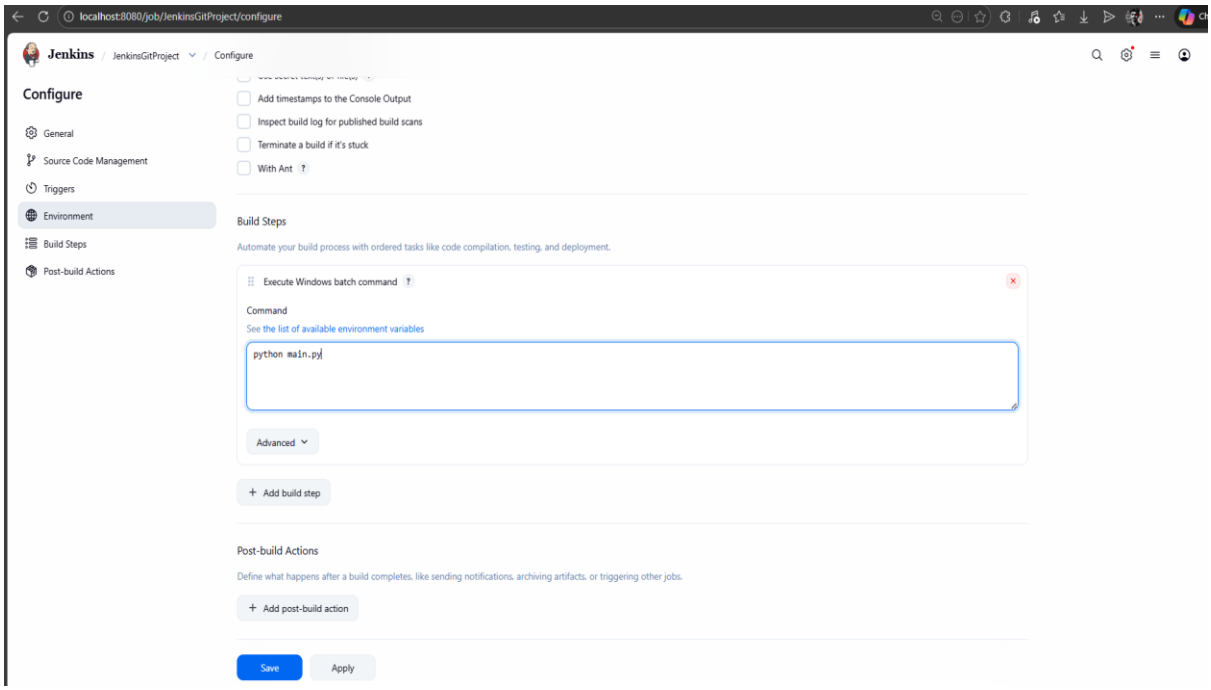


3. Configure **Git in Jenkins**

4. Create Freestyle Job:
    - Source Code Management → Git



5. Add build step:
    - Windows batch / Shell



6. Poll SCM / Webhook trigger

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

- [ ] Trigger builds remotely (e.g., from scripts) ?
- [ ] Build after other projects are built ?
- [ ] Build periodically ?
- [ ] GitHub hook trigger for GITScm polling ?
- [x] Poll SCM ?

Schedule ?

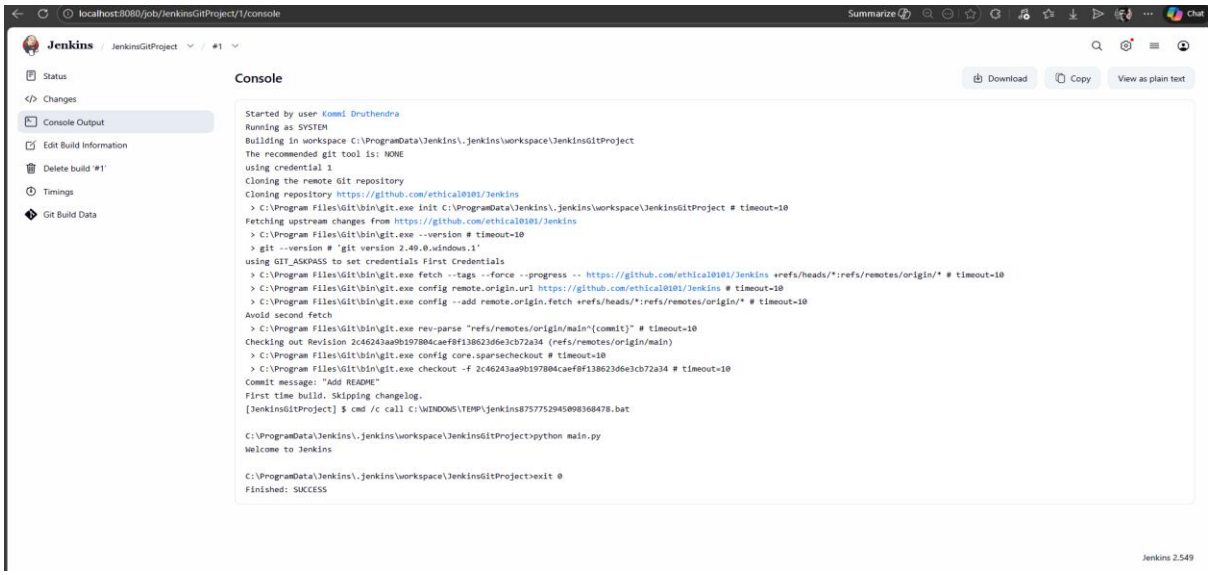
```
* * * * *
```

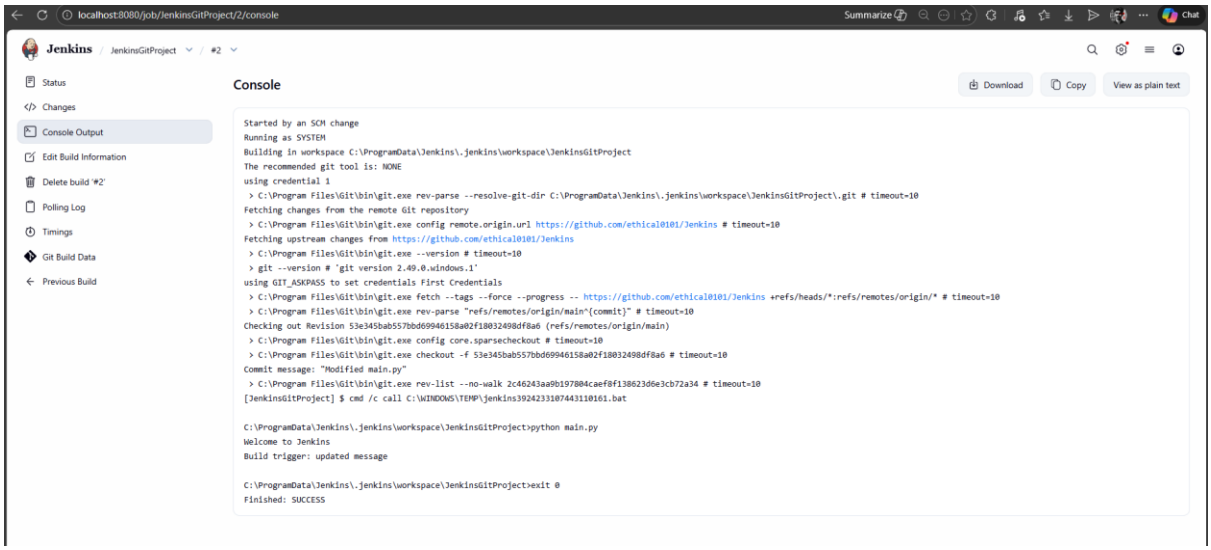
⚠ Do you really mean "every minute" when you say "* * * * *"? Perhaps you meant "H * * * *" to poll once per hour

Would last have run at Thursday, 5 February 2026, 7:25:00 pm India Standard Time; would next run at Thursday, 5 February 2026, 7:26:00 pm India Standard Time.

- [ ] Ignore post-commit hooks ?

7. Build automatically on code change



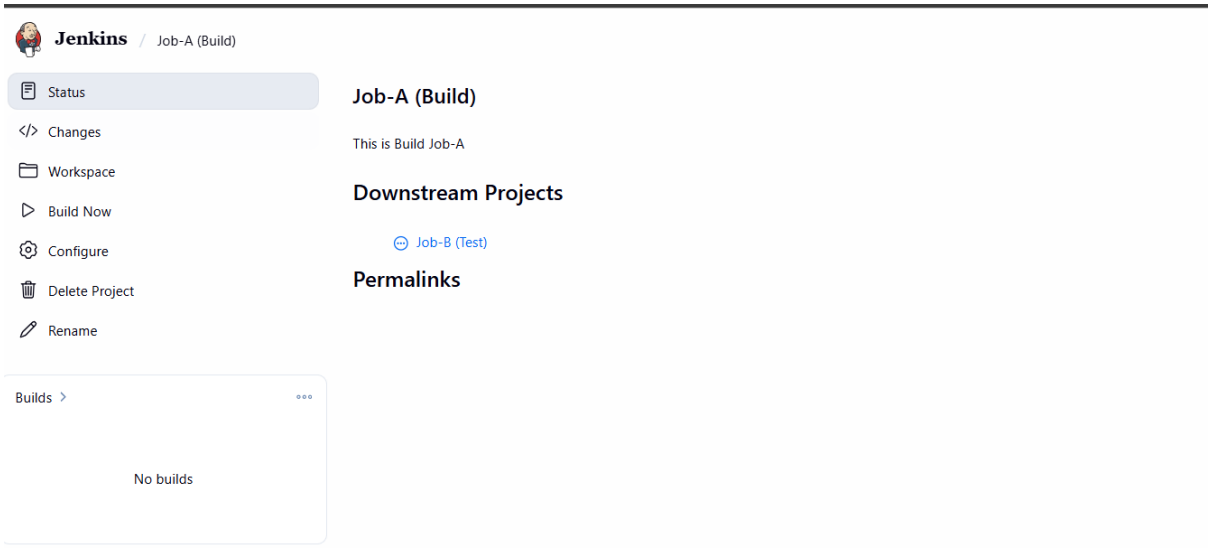


3. Learning Goal

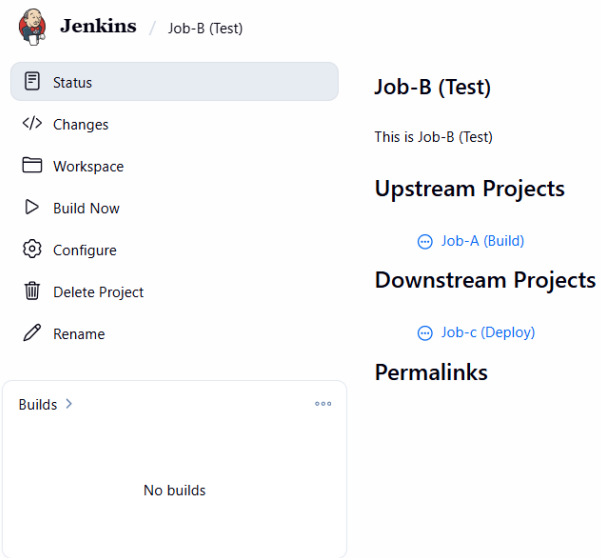
Understand **job flow visually**

Stepwise Tasks

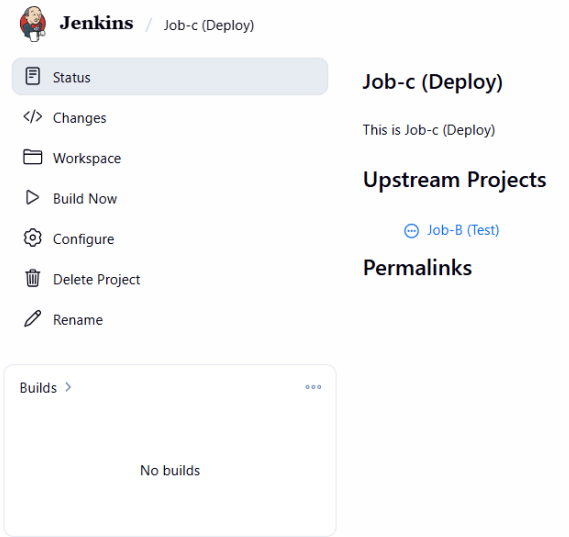
1. Create multiple Freestyle jobs:
    - Job-A (Build)



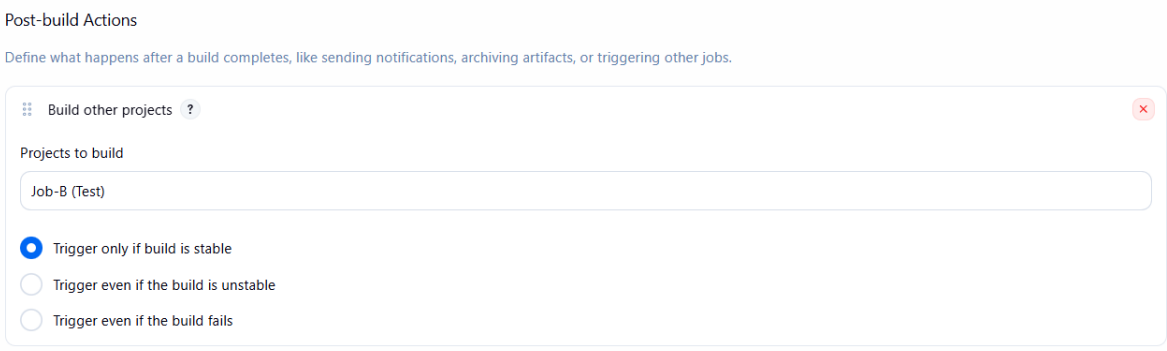
    - Job-B (Test)

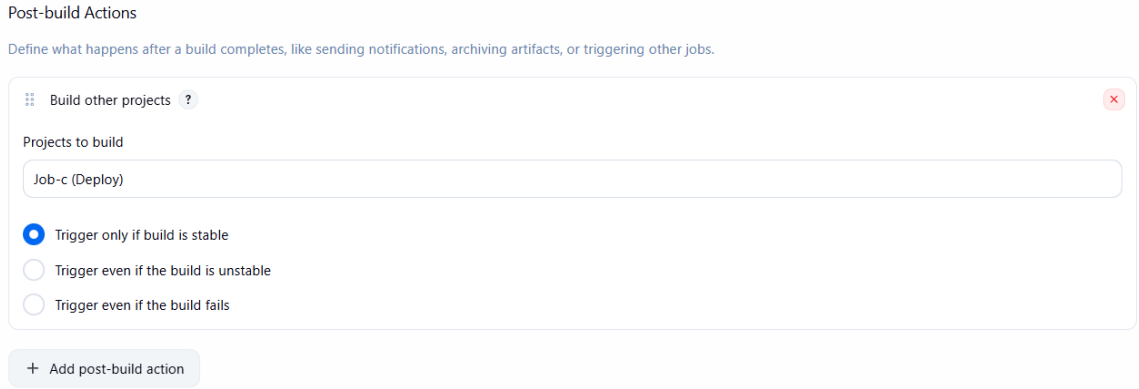


- Job-C (Deploy)

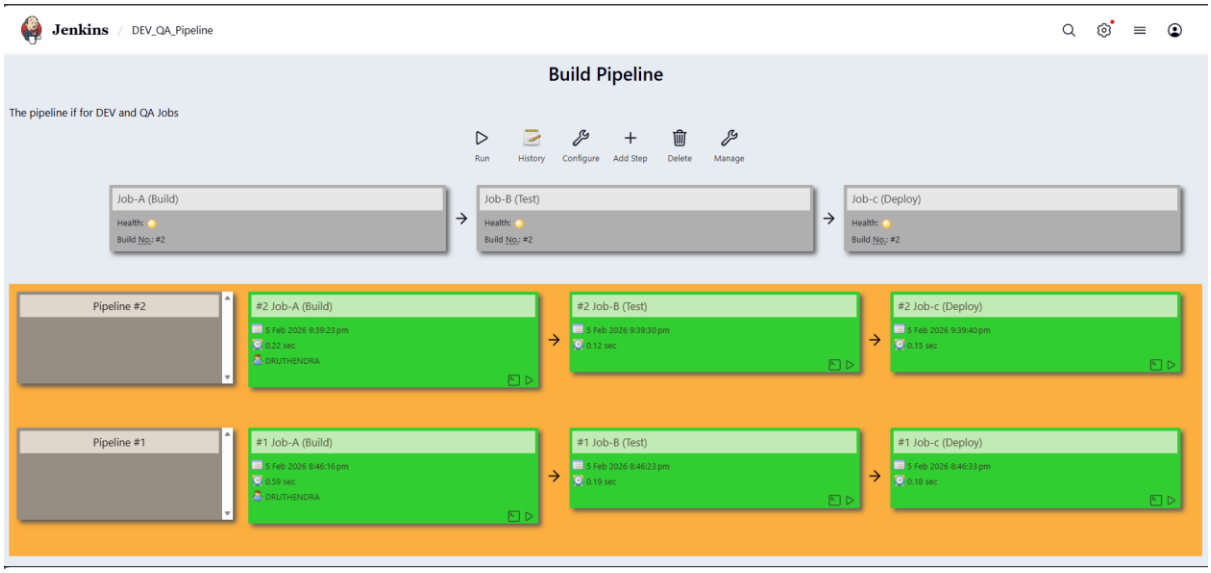


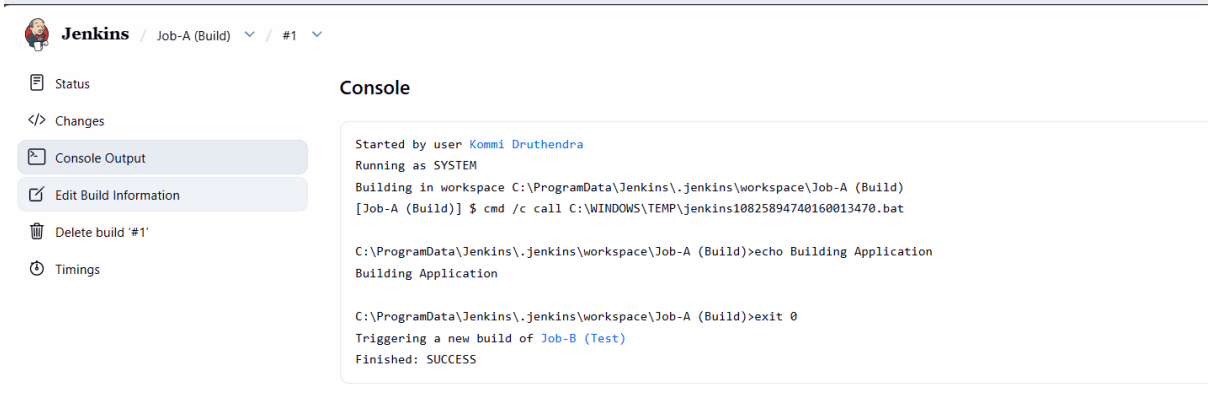
2. Use:
   - Post-build action → Trigger other projects

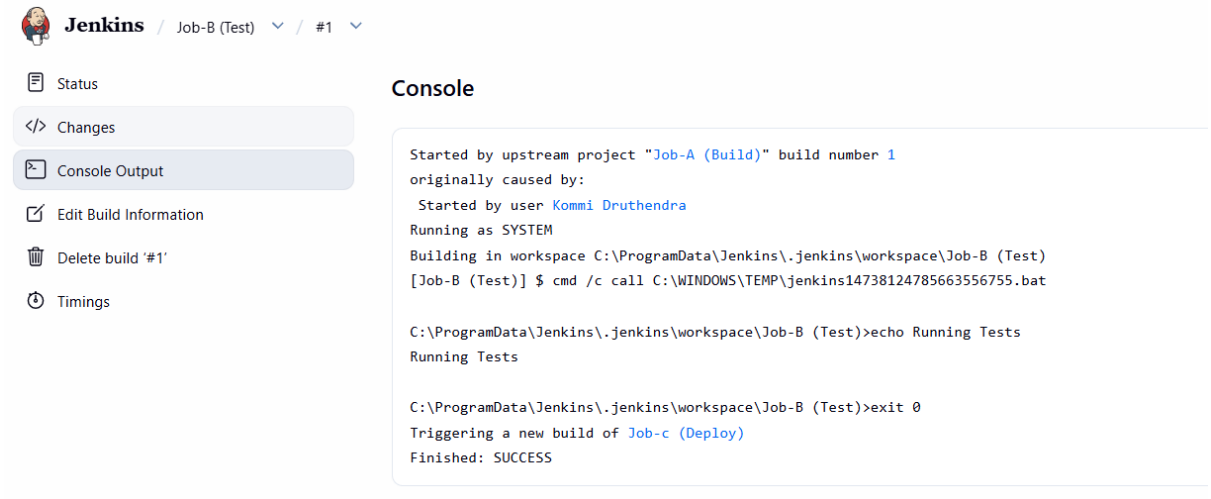


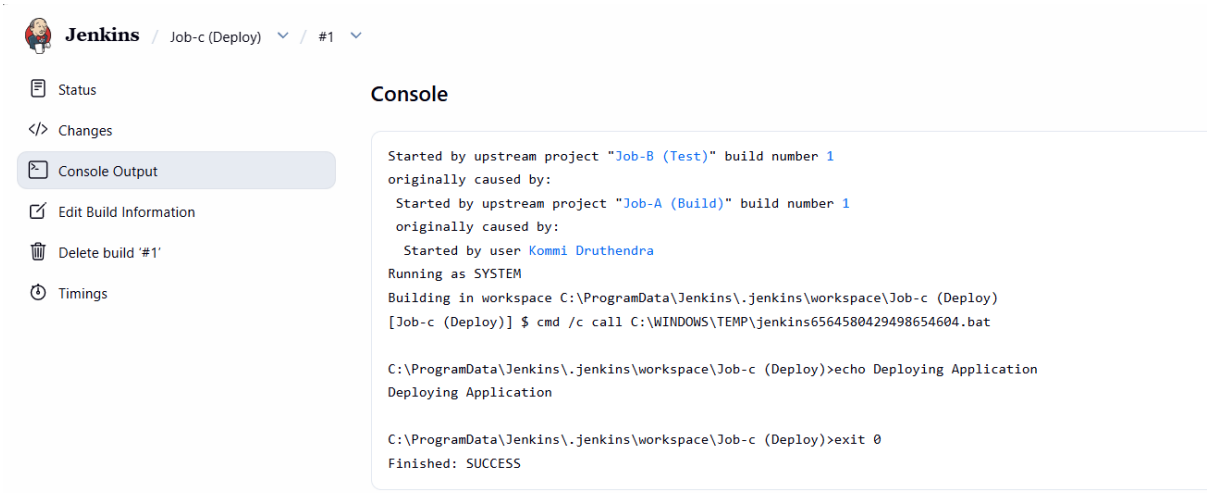


3. View job chaining visually









4. Install **Pipeline Stage View**

4. Learning Goal

Use **Jenkins Pipeline & Jenkinsfile**

Stepwise Tasks

1. What is Jenkins Pipeline?

   A **Jenkins Pipeline** is a suite of automated steps that defines the complete CI/CD process such as:

   - Build
   - Test
   - Deploy

   It allows the entire workflow to be written as code and executed in Jenkins. Pipeline provides:

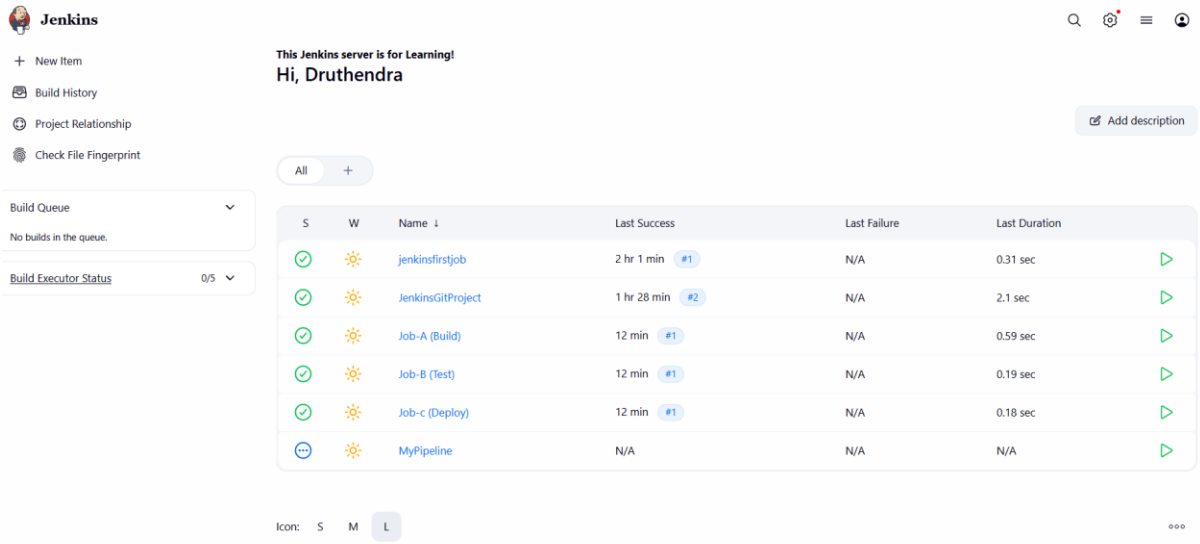
   - Automation of software delivery
   - Stage-wise execution
   - Better visualization
   - Reusability and version control

2. Scripted vs Declarative Pipeline

Feature	Scripted Pipeline	Declarative Pipeline
Syntax	Groovy based	Structured syntax
Complexity	More complex	Simple & readable
Recommended	Advanced users	Beginners & industry
Error handling	Manual	Built-in

Declarative pipeline is recommended because it is easy to understand and maintain.

3. Create **Pipeline Job**



4. Write pipeline in Jenkins UI

```
pipeline {
  agent any

  stages {

    stage('Build') {
      steps {
        echo 'Build Stage'
```

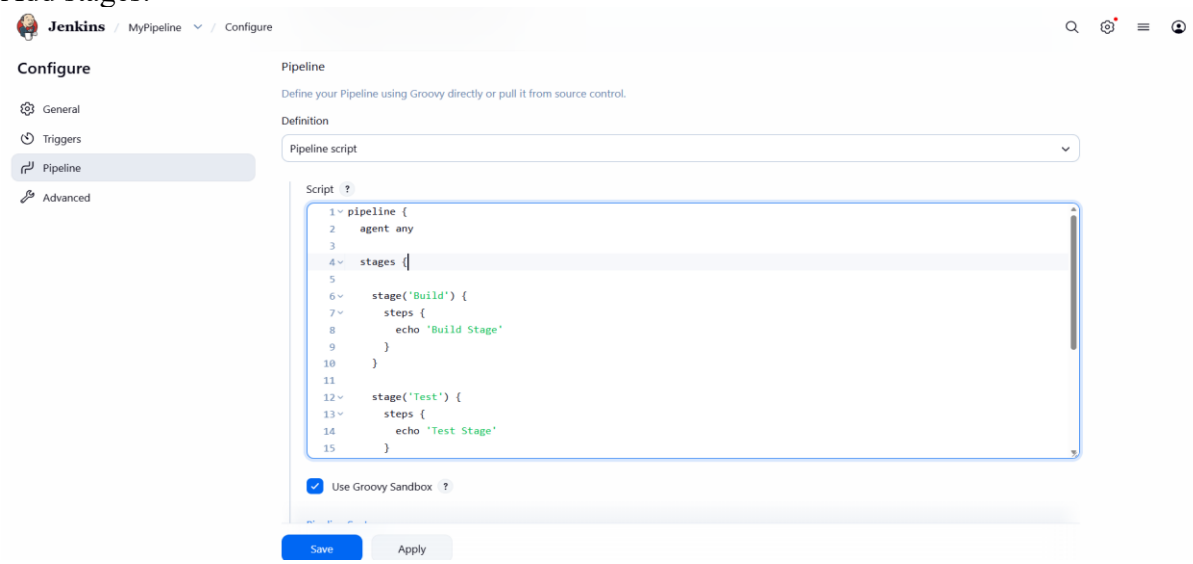
```
    }
  }

  stage('Test') {
    steps {
      echo 'Test Stage'
    }
  }

  stage('Deploy') {
    steps {
      echo 'Deploy Stage'
    }
  }

}
}
```

5. Add stages:



6. Introduce:
    - `agent`
    - `stages`
    - `steps`

```
pipeline {

  agent any



  stages {



    stage('Build') {

      steps {

        echo 'Build Stage'

      }

    }



    stage('Test') {

      steps {

        echo 'Test Stage'

      }

    }



    stage('Deploy') {

      steps {

        echo 'Deploy Stage'

      }

    }
```

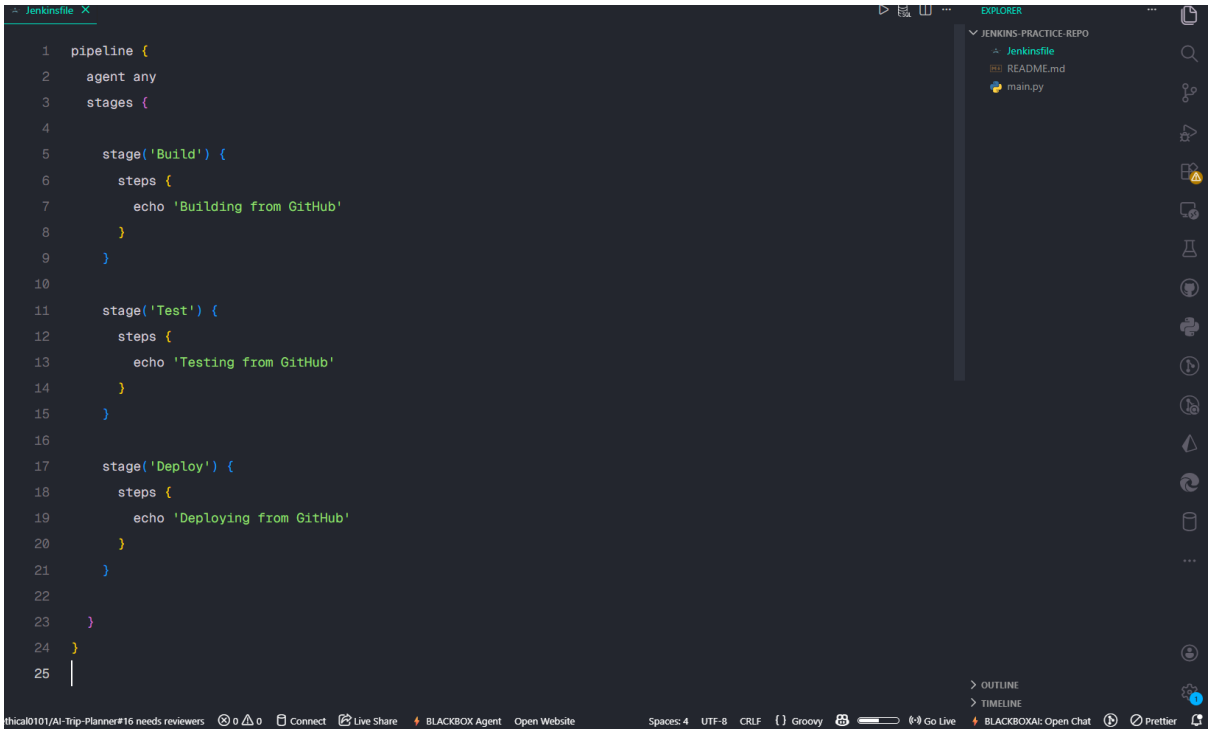
```
  }
}
}
```

5. Learning Goal

Pipeline as code from repository

Stepwise Tasks

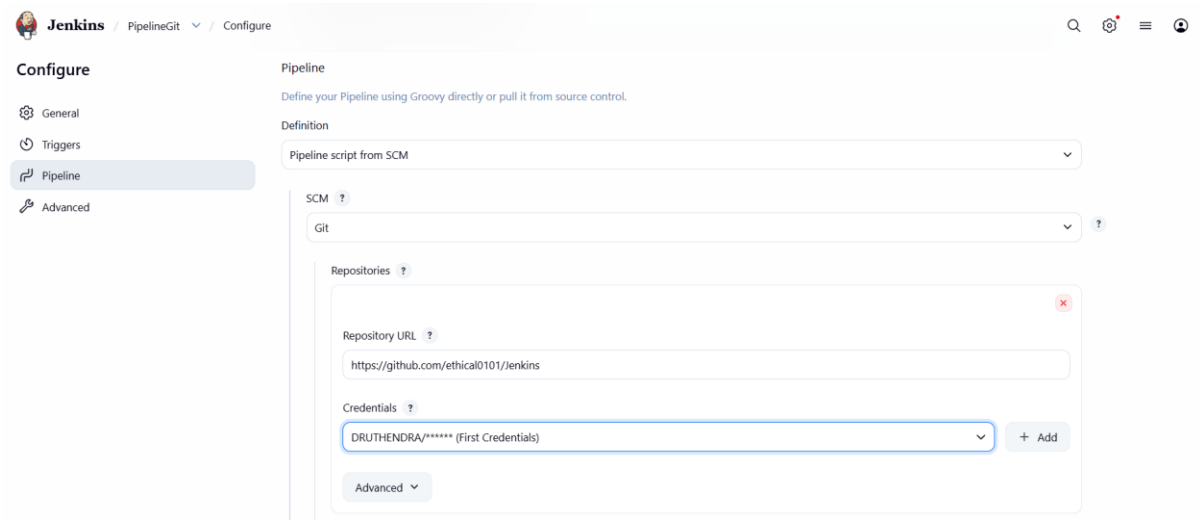
1. Create `Jenkinsfile`



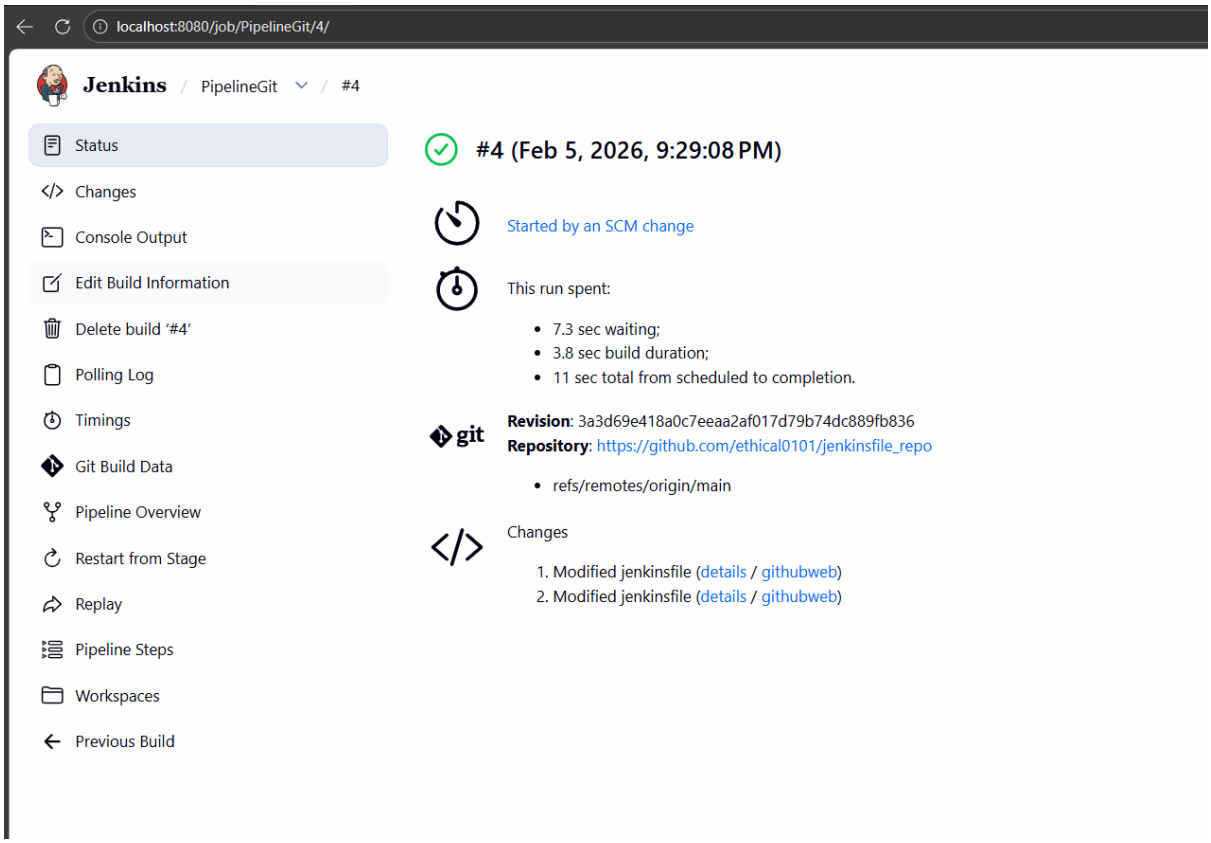
2. Commit Jenkinsfile to GitHub

```
PS E:\Jenkins-Practice-repo> git add .
PS E:\Jenkins-Practice-repo> git commit -m "Modified jenkinsfile"
[main 0968803] Modified jenkinsfile
 1 file changed, 1 insertion(+), 1 deletion(-)
PS E:\Jenkins-Practice-repo> git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 365 bytes | 365.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/ethical0101/Jenkins.git
   8e42b98..0968803  main -> main
```

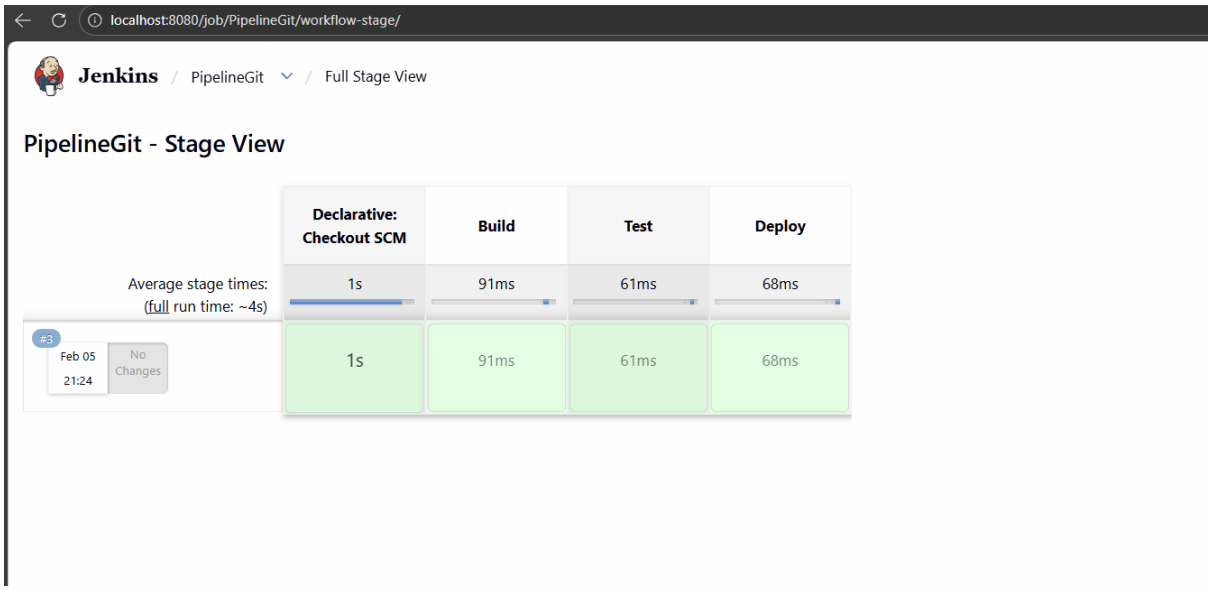
3. Configure Pipeline job:
   - SCM → Git



4. Trigger build automatically



5. Modify Jenkinsfile and rebuild

```
PS E:\jenkinsfile_Repo> git add .
PS E:\jenkinsfile_Repo> git commit -m "Modified jenkinsfile"
[main 4c3d4a5] Modified jenkinsfile
 1 file changed, 1 insertion(+), 1 deletion(-)
PS E:\jenkinsfile_Repo> git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 288 bytes | 288.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/ethical0101/jenkinsfile_repo.git
   1fab71e..4c3d4a5  main -> main
```

6. View Stage View